

Exam Date & Time: 10-May-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech IV Semester

End Semester Examination - April / May -2025

Electromagnetic Waves [ECM-2222]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Three point charges $Q_1 = -1 \mu\text{C}$, $Q_2 = -2 \mu\text{C}$ and $Q_3 = -3 \mu\text{C}$ are placed at the corners of an equilateral triangle of side 1m . Find the magnitude of the electric field intensity at the point bisecting the line joining Q_1 and Q_2 . (4)
 - A)
 - B) State and prove Gauss law in point form for electrostatics. (4)
 - C) Find the divergence of $\mathbf{A}$ at $P(5, \pi/2, 1)$ where $\mathbf{A} = rz \sin \phi \mathbf{a}_r + 3rz^2 \cos \phi \mathbf{a}_\phi$ (2)
- 2) State and prove Ampere's circuital law. (5)
 - A)
 - B) Using Biot-Savart's law, derive the expression for magnetic field intensity at point due to current in a straight conductor of infinite length. (5)
- 3) Given the potential field $V = 3x^2yz + ky^3z$ volts,
 - a. Find k if potential field satisfies Laplace equation (3)
 - b. Find the electric field intensity at $(1, 2, 3)$
 - A)
 - B) If the magnetic field intensity in a region is $\mathbf{H} = (3y - 2)\mathbf{a}_z + 2x\mathbf{a}_y$, find the current density at the origin. (2)

- C) List the Maxwell's equations for static fields. Mention the representation of all the variables in the equation. (5)
- 4) Explain the reflection of plane waves at the surface of dielectrics at normal incidence. Hence derive the expression for reflection and transmission coefficients. (5)
- A)
- B) A 10 GHz plane wave travelling in a free space has an amplitude of E as $E_x = 10 \text{ V/m}$. Calculate β, η, v, λ and the amplitude of H . (5)
- 5) a. Write a note on loss-less and low-loss transmission lines and derive the expression for propagation constant in both the cases. (5)
- b. With a neat diagram, write a detailed note on the constant reactance circles in a complex reflection coefficient plane
- A)
- B) A generator of 2 volts, 1000 Hz supplies power to a 100 km open wire terminated in Z_0 having the following parameters
- $R = 10 \Omega/\text{km}$
- $G = 0.8 \mu \text{ S}/\text{km}$
- $L = 0.004 \text{ H}/\text{km}$
- $C = 0.008 \mu\text{F}/\text{km}$ (5)
- Calculate $Z_0, \gamma, \alpha, \beta$ and λ

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